IEOR 221
Homework Solutions

Problem 1

(a) Standard Deviations of Stocks A and B

Under the single-index model,

$$\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma_{ei}^2.$$

Stock A:

$$\sigma_A^2 = (0.8)^2(0.22)^2 + (0.30)^2 = 0.64(0.0484) + 0.09 = 0.030976 + 0.09 = 0.120976.$$

$$\sigma_A = \sqrt{0.120976} = 0.3478 \approx \boxed{34.8\%}.$$

Stock B:

$$\sigma_B^2 = (1.2)^2(0.22)^2 + (0.40)^2 = 1.44(0.0484) + 0.16 = 0.069696 + 0.16 = 0.229696.$$

$$\sigma_B = \sqrt{0.229696} = 0.4793 \approx \boxed{47.9\%}.$$

(b) Portfolio Expected Return, Beta, Idiosyncratic Risk, Total Risk

Weights:

$$w_A = 0.30, \quad w_B = 0.45, \quad w_f = 0.25.$$

(i) Expected Return

$$E[R_p] = 0.30(0.13) + 0.45(0.18) + 0.25(0.08).$$

$$E[R_p] = 0.039 + 0.081 + 0.020 = \boxed{0.14 = 14\%}.$$

(ii) Portfolio Beta

$$\beta_p = 0.30(0.8) + 0.45(1.2) + 0.25(0) = 0.24 + 0.54 = \boxed{0.78}.$$

(iii) Idiosyncratic Standard Deviation

$$\sigma_{ep}^2 = w_A^2\sigma_{eA}^2 + w_B^2\sigma_{eB}^2 = 0.09(0.09) + 0.2025(0.16).$$

$$\sigma_{ep}^2 = 0.0081 + 0.0324 = 0.0405.$$

$$\sigma_{ep} = \sqrt{0.0405} = 0.2012 \approx \boxed{20.1\%}.$$

(iv) Total Standard Deviation

Systematic variance:

$$\sigma_{\text{sys},p}^2 = \beta_p^2\sigma_M^2 = (0.78)^2(0.22)^2 = 0.6084(0.0484) = 0.02944.$$

Total variance:

$$\sigma_p^2 = 0.02944 + 0.0405 = 0.06994.$$

$$\sigma_p = \sqrt{0.06994} = 0.26455 \approx \boxed{26.5\%}.$$

Problem 2

Given:

$$r_f = 6\%, \qquad E[R_M] = 16\%, \qquad E[R_M] - r_f = 10\%.$$

(a) Expected Stock Price at Year End

For $\beta = 1.2$,

$$E[R] = r_f + \beta(E[R_M] - r_f) = 0.06 + 1.2(0.10) = 0.18 = 18\%.$$

Use

$$E[R] = \frac{P_1 + D_1 - P_0}{P_0}.$$

$$0.18 = \frac{P_1 + 6 - 50}{50} \Rightarrow P_1 + 6 - 50 = 9 \Rightarrow P_1 = \boxed{53}.$$

(b) Incorrect Beta and Overpayment

Firm generates \$100,000 every year forever.

$$r_{\text{true}} = 0.06 + 1(0.10) = 0.16.$$
$$r_{\text{guess}} = 0.06 + 0.5(0.10) = 0.11.$$

True value:

$$PV_{\text{true}} = \frac{100{,}000}{0.16} = 625{,}000.$$

Investor's (wrong) valuation:

$$PV_{\text{guess}} = \frac{100{,}000}{0.11} = 909{,}091.$$

Overpayment:

$$PV_{\text{guess}} - PV_{\text{true}} = 909{,}091 - 625{,}000 = \boxed{284{,}091}.$$

(c) Beta of a Stock with 4% Expected Return

$$0.04 = 0.06 + \beta(0.10) \Rightarrow -0.02 = 0.10\beta \Rightarrow \beta = \boxed{-0.2}.$$

Problem 3

Given:

$$E[R_F] = 14\%, \qquad \beta_F = 0.8, \qquad r_f = 5\%, \qquad E[R_M] = 15\%.$$

Risk premium:

$$E[R_M] - r_f = 10\%.$$

(a) Should You Invest in the Fund?

CAPM-required return:

$$E[R_{\text{CAPM}}] = 0.05 + 0.8(0.10) = 0.13 = 13\%.$$

Fund offers 14%, so

$$\alpha = 0.14 - 0.13 = \boxed{1\%}.$$

Since $\alpha > 0$, the fund lies above the SML.

$$\boxed{\text{Yes, the fund is attractive.}}$$

(b) Passive Portfolio with Same Beta

Let w_M be market weight. Beta of passive portfolio:

$$\beta_p = w_M = 0.8.$$

Thus

$$w_M = 0.8, \qquad w_f = 0.2.$$

Expected return:

$$E[R_p] = 0.8(0.15) + 0.2(0.05) = 0.12 + 0.01 = 0.13.$$

$$\boxed{\text{80\% market, 20\% risk-free; same beta but lower return (13\%).}}$$